

Harder Inequalities (進階不等式)

1. Permutation Inequality (排序不等式)

For any two sequences of real numbers $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$, both arranged in ascending order of magnitude,

$$a_1b_n + a_2b_{n-1} + a_3b_{n-2} + \dots + a_nb_1 \leq a_1b_\alpha + a_2b_\beta + a_3b_\gamma + \dots + a_nb_\lambda \leq a_1b_1 + a_2b_2 + a_3b_3 + \dots + a_nb_n,$$

where $\alpha, \beta, \gamma, \dots, \lambda$ is any arbitrary permutation of the numbers $1, 2, 3, \dots, n$.

The equality holds if and only if $a_1 = a_2 = a_3 = \dots = a_n$ or $b_1 = b_2 = b_3 = \dots = b_n$.

In other words, 逆序和 $\leq$ 亂序和 $\leq$ 順序和

Proof: Since $(a_n - a_i)(b_n - b_\lambda) \geq 0$, hence $a_ib_\lambda + a_nb_n \geq a_ib_n + a_nb_\lambda$

Hence if we consider a sequence of n terms $a_1b_\alpha + a_2b_\beta + a_3b_\gamma + \dots + a_nb_\lambda$ and replace $a_ib_n + a_nb_\lambda$ by $a_ib_\lambda + a_nb_n$, then the sum will be increased.

Next we repeat with the first $n-1$ terms and then obtain the last two terms $a_{n-1}b_{n-1} + a_nb_n$.

Repeat the procedure until we get $a_1b_1 + a_2b_2 + a_3b_3 + \dots + a_nb_n$, which is largest sum.

The proof of the other part is similar.

Example 1: The 900 students of a certain school belong to 4 different houses, with 230, 275, 220, 175 students respectively. The school will provide a different type of T-shirt to the members of each house. There are four kinds of T-shirts, with unit prices \$120, \$90, \$115, and \$130 respectively. What is the minimum cost of providing T-shirts to all students in the school?

Example 2: 10 men in a queue are waiting for filling their buckets with water. The buckets are of different sizes, and the i^{th} man needs t_i minutes to fill his bucket. How should be the order be arranged so that the total time spent by the 10 men on waiting will be least.

Example 3: Prove that for any real numbers a, b and c ,

$$(i) a^2 + b^2 \geq 2ab. (ii) a^2 + b^2 + c^2 \geq ab + bc + ca (iii) (a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$$

Example 4: For any positive numbers $a_1, a_2, \dots, a_n$, prove that

$$\frac{(a_1)^2}{a_2} + \frac{(a_2)^2}{a_3} + \frac{(a_3)^2}{a_4} + \dots + \frac{(a_{n-1})^2}{a_n} + \frac{(a_n)^2}{a_1} \geq a_1 + a_2 + a_3 + \dots + a_n.$$

Example 5: Let a, b, c, d be real numbers such that $0 < a \leq b \leq c \leq d$, and $a + b + c + d \geq 1$. Prove that $a^2 + 3b^2 + 5c^2 + 7d^2 \geq 1$.

Example 6: Let a, b, c be the lengths of the sides of $\triangle ABC$. Prove that

$$a^2(b+c-a) + b^2(c+a-b) + c^2(a+b-c) \leq 3abc.$$

Example 7: For any two sequences of real numbers $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$, both arranged in ascending order of magnitude, suppose $i_1, i_2, \dots, i_n$ and $j_1, j_2, \dots, j_n$ are two arbitrary arrangements of $1, 2, \dots, n$, prove that

$$\sum_{r=1}^n \sum_{s=1}^n \frac{a_{i_r} b_{j_s}}{r+s} \geq \sum_{r=1}^n \sum_{s=1}^n \frac{a_r b_s}{r+s}.$$

Example 8: (17th IMO (1975) No. 1) Let x_i, y_i ($i=1, 2, \dots, n$) be real numbers such that $x_1 \geq x_2 \geq \dots \geq x_n$ and $y_1 \geq y_2 \geq \dots \geq y_n$.

Prove that if $z_1, z_2, \dots, z_n$ is any permutation of $y_1, y_2, \dots, y_n$,

$$\text{then } \sum_{i=1}^n (x_i - y_i)^2 \leq \sum_{i=1}^n (x_i - z_i)^2$$

2. Chebyshev Inequality (切比雪夫不等式)

(Pafnuty Lvovich Chebyshev, Russian mathematician, 1821-1894)

For any two sequences of real numbers $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$, both arranged in ascending order of magnitude,

$$\frac{a_1 b_n + a_2 b_{n-1} + \dots + a_n b_1}{n} \leq \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right) \left(\frac{b_1 + b_2 + \dots + b_n}{n} \right) \leq \frac{a_1 b_1 + a_2 b_2 + \dots + a_n b_n}{n}$$

$$\text{In other words, } \frac{\sum_{i=1}^n a_i b_{n+1-i}}{n} \leq \left(\frac{\sum_{i=1}^n a_i}{n} \right) \left(\frac{\sum_{i=1}^n b_i}{n} \right) \leq \frac{\sum_{i=1}^n a_i b_i}{n}$$

Equality holds only if $a_1 = a_2 = \dots = a_n$, or $b_1 = b_2 = \dots = b_n$.

Proof: (Sum up the following inequalities:

$$a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots + a_n b_n \leq a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots + a_n b_n$$

$$a_1 b_2 + a_2 b_3 + a_3 b_4 + \dots + a_n b_1 \leq a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots + a_n b_n$$

$$a_1 b_3 + a_2 b_4 + a_3 b_5 + \dots + a_n b_2 \leq a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots + a_n b_n$$

.....

$$a_1 b_n + a_2 b_1 + a_3 b_2 + \dots + a_n b_{n-1} \leq a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots + a_n b_n$$

Finally, divide the sum on both sides by n^2 .

$$\text{We have } \left(\frac{a_1 + a_2 + \dots + a_n}{n} \right) \left(\frac{b_1 + b_2 + \dots + b_n}{n} \right) \leq \frac{a_1 b_1 + a_2 b_2 + \dots + a_n b_n}{n}$$

The other half of the inequality can be proved in a similar way.

Example 9: Prove that for any sequence $\{a_i\}$ of positive real numbers, and any positive integer m ,

$$\frac{\sum_{i=1}^n a_i^m}{n} \geq \left(\frac{\sum_{i=1}^n a_i}{n} \right)^m, \text{ hence prove that } 1^3 + 2^3 + \dots + n^3 > \frac{n(n+1)^3}{8}.$$

Example 10: Given that $a \leq b \leq c \leq d \leq e$, and that $a + b + c + d + e = 1$, prove that

$$ad + ae + bc + be + cd \leq \frac{1}{5}.$$

Example 11: Given that $x_1, x_2, \dots, x_n$ are positive real numbers and $k \geq 1$. Prove that

$$\left(\sum_{i=1}^n \frac{1}{1+x_i} \right) \left(\sum_{i=1}^n x_i \right) \leq \left(\sum_{i=1}^n \frac{x_i^{k+1}}{1+x_i} \right) \left(\sum_{i=1}^n \frac{1}{x_i^k} \right)$$

3. Jensen's Inequality (琴生不等式) (Johan Jensen, Danish mathematician, 1859-1925)

A function $f(x)$ is **convex** ($\cup$) on an interval I if for every $x_1, x_2 \in I$,

$$f\left(\frac{x_1 + x_2}{2}\right) \leq \frac{f(x_1) + f(x_2)}{2}$$

It is said to be **strictly convex** on I if the equality holds only when $x_1 = x_2$.

A necessary condition for $f(x)$ to be **strictly convex** in I is that $f'(x) > 0$ in I .

The following are examples of **strictly convex** functions:

- (i) $f(x) = x^p$ on $[0, \infty)$ for $p > 1$, (ii) $f(x) = a^x$ on $(-\infty, \infty)$ for $a > 1$, (iii) $f(x) = \cos x$ on $[\frac{\pi}{2}, \frac{3\pi}{2}]$.

On the other hand, the function $f(x)$ is **concave** ($\cap$) on an interval I if for every $x_1, x_2 \in I$,

$$f\left(\frac{x_1 + x_2}{2}\right) \geq \frac{f(x_1) + f(x_2)}{2}$$

It is said to be **strictly concave** on I if the equality holds only when $x_1 = x_2$.

A necessary condition for $f(x)$ to be **strictly concave** in I is that $f'(x) < 0$ in I .

The following are examples of **strictly concave** functions:

- (i) $f(x) = \sin x$ on $[0, \pi]$, (ii) $f(x) = x^p$ on $[0, \infty)$ for $0 < p < 1$, (iii) $f(x) = \tan x$ on $(\frac{\pi}{2}, \pi]$.

Theorem

- (i) If $f(x)$ is a convex function on an interval I , and $x_1, x_2, x_3, \dots, x_n \in I$, then

$$f\left(\frac{x_1 + x_2 + \dots + x_n}{n}\right) \leq \frac{f(x_1) + f(x_2) + \dots + f(x_n)}{n}$$

The equality holds if and only if $x_1 = x_2 = x_3 = \dots = x_n$.

- (ii) If $f(x)$ is a convex function on an interval I , $x_1, x_2, x_3, \dots, x_n \in I$, and $q_1, q_2, q_3, \dots, q_n$ are n positive real numbers such that $q_1 + q_2 + q_3 + \dots + q_n = 1$, then

$$f(q_1x_1 + q_2x_2 + q_3x_3 + \dots + q_nx_n) \leq [q_1f(x_1) + q_2f(x_2) + q_3f(x_3) + \dots + q_nf(x_n)]$$

The equality holds if and only if $x_1 = x_2 = x_3 = \dots = x_n$.

- (iii) If $f(x)$ is a convex function on an interval I , $x_1, x_2, x_3, \dots, x_n \in I$, and $p_1, p_2, p_3, \dots, p_n$ are n positive real numbers, then

$$f\left(\frac{p_1x_1 + p_2x_2 + \dots + p_nx_n}{p_1 + p_2 + \dots + p_n}\right) \leq \frac{p_1f(x_1) + p_2f(x_2) + \dots + p_nf(x_n)}{p_1 + p_2 + \dots + p_n}$$

The equality holds if and only if $x_1 = x_2 = x_3 = \dots = x_n$.

Proof:

- (i) By Mathematical Induction on n

- (ii) For $n = 2$, refer to the diagram.

Let q_1, q_2 be real numbers such that $q_1 + q_2 = 1$,
and $HL : LK = q_2 : q_1$, then since $\frac{x-x_1}{x_2-x_1} = \frac{q_2}{q_1}$,
giving $x = q_1x_1 + q_2x_2$,

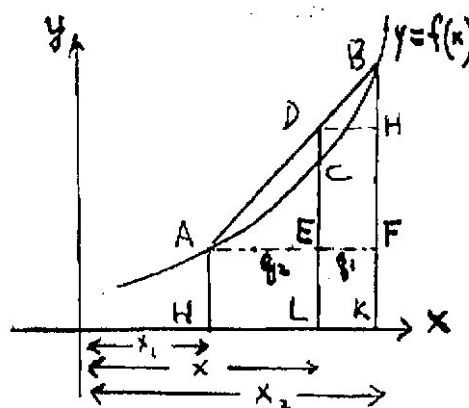
Using similar triangles, $\frac{BH}{DE} = \frac{q_2}{q_1} \Rightarrow \frac{DL - f(x_1)}{f(x_2) - DL} = \frac{q_2}{q_1}$

Hence $DL = q_1f(x_1) + q_2f(x_2)$

But $CL = f(q_1x_1 + q_2x_2)$

Obviously $CL < DL$

Hence $f(q_1x_1 + q_2x_2) < q_1f(x_1) + q_2f(x_2)$



Next, assume that the proposition is true when $n = k$, This means that for any k points $x_1, x_2, x_3, \dots, x_k \in I$, and k positive numbers $q_1, q_2, q_3, \dots, q_k$ such that $q_1 + q_2 + q_3 + \dots + q_k = 1$, then $f(q_1 x_1 + q_2 x_2 + q_3 x_3 + \dots + q_k x_k) \leq [q_1 f(x_1) + q_2 f(x_2) + q_3 f(x_3) + \dots + q_k f(x_k)]$

Hence for any $k+1$ points $x_1, x_2, x_3, \dots, x_k, x_{k+1} \in I$, and $k+1$ positive numbers $q_1, q_2, q_3, \dots, q_k, q_{k+1}$ such that $q_1 + q_2 + q_3 + \dots + q_k + q_{k+1} = 1$, then set

$$x_k' = \frac{q_k x_k + q_{k+1} x_{k+1}}{q_k + q_{k+1}} \quad \text{and} \quad q_k' = q_k + q_{k+1}$$

From our assumption

$$f(q_1 x_1 + q_2 x_2 + q_3 x_3 + \dots + q_k' x_k') \leq [q_1 f(x_1) + q_2 f(x_2) + q_3 f(x_3) + \dots + q_k' f(x_k')]$$

Hence $f(q_1 x_1 + q_2 x_2 + q_3 x_3 + \dots + q_k x_k + q_{k+1} x_{k+1})$

$$\leq q_1 f(x_1) + q_2 f(x_2) + q_3 f(x_3) + \dots + (q_k + q_{k+1}) f\left[\left(\frac{q_k}{q_k + q_{k+1}}\right)x_k + \left(\frac{q_{k+1}}{q_k + q_{k+1}}\right)x_{k+1}\right]$$

$$\leq q_1 f(x_1) + q_2 f(x_2) + q_3 f(x_3) + \dots + (q_k + q_{k+1}) \left[\left(\frac{q_k}{q_k + q_{k+1}}\right)f(x_k) + \left(\frac{q_{k+1}}{q_k + q_{k+1}}\right)f(x_{k+1}) \right]$$

$$= [q_1 f(x_1) + q_2 f(x_2) + q_3 f(x_3) + \dots + q_k f(x_k) + q_{k+1} f(x_{k+1})]$$

Hence the proposition is also true for $n=k+1$.

By the Principle of Mathematical Induction the proposition is true for any positive integer n .

(iii) For any set of positive integers $p_1, p_2, \dots, p_n$, we can set $q_i = \frac{p_i}{p_1 + p_2 + \dots + p_n}$, then

$$\sum q_i = 1, \quad \text{and} \quad f\left(\frac{\sum_{i=1}^n p_i x_i}{\sum_{i=1}^n p_i}\right) \leq \frac{\sum_{i=1}^n p_i f(x_i)}{\sum_{i=1}^n p_i}$$

Example 12: Prove that if A, B, C are the angles of a triangle, then $\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$.

Example 13: Given that $f(x) = \log_e(\sin x)$ is a concave function in $(0, \pi)$, prove that if A, B, C

$$\text{are the angles of a triangle, then } \sin A \sin B \sin C \leq \frac{3\sqrt{3}}{8}$$

Example 14: Prove that for any integer $n \geq 3$,

$$\sqrt{9n+8} < \sqrt{n} + \sqrt{n+1} + \sqrt{n+2} < 3\sqrt{n+1}.$$

Example 15: Use Jensen's Inequality to prove $G.M. \leq A.M. \leq R.M.S$

Example 16: Use Jensen's Inequality to prove Power Mean Inequality (幂平均不等式)

For any n ($n > 1$) non-negative real numbers $a_1, a_2, a_3, \dots, a_n$, and real numbers $s > t > 0$,

$$\left(\frac{a_1^s + a_2^s + \dots + a_n^s}{n}\right)^{\frac{1}{s}} \geq \left(\frac{a_1^t + a_2^t + \dots + a_n^t}{n}\right)^{\frac{1}{t}}$$

The equality holds if and only if $a_1 = a_2 = a_3 = \dots = a_n$.

Proof: Consider $f(x) = x^k$ ($k > 1$), which is a strictly convex function on $(0, \infty)$,

$$\text{Hence} \quad \left(\frac{\sum_{i=1}^n x_i}{n}\right)^k \leq \frac{\sum_{i=1}^n x_i^k}{n}. \quad \text{Now put } k = \frac{\alpha}{\beta}, \text{ where } \alpha > \beta, \text{ and } x_i = a_i^\beta$$

Then
$$\left(\frac{\sum_{i=1}^n a_i^\beta}{n} \right)^{\frac{\alpha}{\beta}} \leq \left(\frac{\sum_{i=1}^n a_i^{\beta(\frac{\alpha}{\beta})}}{n} \right), \text{ that is } \left(\frac{\sum_{i=1}^n x_i^\beta}{n} \right)^{\frac{1}{\beta}} \leq \left(\frac{\sum_{i=1}^n x_i^\alpha}{n} \right)^{\frac{1}{\alpha}},$$

which is called the Power Mean Inequality.

The equality holds only when $a_1 = a_2 = a_3 = \dots = a_n$

In conclusion, A.M. $\leq$ R.M.S $\leq \left(\frac{\sum_{i=1}^n x_i^3}{n} \right)^{\frac{1}{3}} \leq \left(\frac{\sum_{i=1}^n x_i^4}{n} \right)^{\frac{1}{4}} \leq \dots$ etc.

Example 17: Let $a_1, a_2, \dots, a_n$ be n positive real numbers and α, β be positive integers.

Prove that
$$\left(\frac{\sum_{i=1}^n a_i^{\alpha+\beta}}{n} \right) \geq \left(\frac{\sum_{i=1}^n a_i^\alpha}{n} \right) \left(\frac{\sum_{i=1}^n a_i^\beta}{n} \right)$$

Deduce that (i) $n(a_1^5 + a_2^5 + \dots + a_n^5) \geq (a_1^2 + a_2^2 + \dots + a_n^2)(a_1^3 + a_2^3 + \dots + a_n^3)$,

(ii) for any $x, y, z > 0$, $x^5 + y^5 + z^5 \leq x^5 \sqrt{\frac{x^2}{yz}} + y^5 \sqrt{\frac{y^2}{zx}} + z^5 \sqrt{\frac{z^2}{xy}}$.

Example 18: Given that $f(x) = x \ln x$ is a convex function for $x > 0$. If a, b are distinct

positive real numbers, prove that $a^a b^b > \left(\frac{a+b}{2} \right)^{a+b}$.

Example 19: Given that $f(x) = \log_e \left(\frac{\sin x}{x} \right)$ is a concave function for $0 < x < \pi$,

suppose $0 < x_i < \pi$ for $i = 1, 2, \dots, n$, and $m = \frac{x_1 + x_2 + \dots + x_n}{n}$.

Prove that
$$\left(\frac{\sin x_1}{x_1} \right) \left(\frac{\sin x_2}{x_2} \right) \dots \left(\frac{\sin x_n}{x_n} \right) \leq \left(\frac{\sin m}{m} \right)^n$$
.

Example 20: Let a, b be real numbers, and r a positive rational number. Prove that

- (i) If $r \leq 1$, then $|a + b|^r \leq |a|^r + |b|^r$
- (ii) If $r > 1$, then $|a + b|^r \leq 2^{r-1} (|a|^r + |b|^r)$.

Proof: We shall only prove for the case $a > 0$ and $b > 0$. (For the case $ab < 0$, then (i) and (ii) are both obvious. For the case $ab = 0$, equality holds. For the case $a < 0$ and $b < 0$, since $|a| = -a$, $|b| = -b$ and $|a+b| = -a-b$, the proof is similar to the case $a > 0$ and $b > 0$.)

(i) If $r \leq 1$, then $(a + b)^r = a(a + b)^{r-1} + b(a + b)^{r-1} \leq a^r + b^r$.

(because for $0 < r \leq 1$, $g(x) = x^{r-1}$ is a decreasing function.)

(ii) If $r > 1$, let $f(x) = x^r$, then $f'(x) = r x^{r-1}$, $f''(x) = r(r-1) x^{r-2} > 0$ for $x > 0$.

Hence $f(x)$ is a convex function for $x > 0$.

$$f\left(\frac{a+b}{2}\right) \leq \frac{1}{2} f(a) + \frac{1}{2} f(b) \quad \text{that is} \quad \left(\frac{a+b}{2}\right)^r \leq \frac{1}{2} (a^r + b^r)$$

Hence $(a + b)^r \leq 2^{r-1} (a^r + b^r)$, which implies $|a + b|^r \leq 2^{r-1} (|a|^r + |b|^r)$.

Exercise

Ex. 1: For any positive real numbers a, b and c , prove that

$$a + b + c \leq \frac{a^2 + b^2}{2c} + \frac{b^2 + c^2}{2a} + \frac{c^2 + a^2}{2b} \leq \frac{a^3}{bc} + \frac{b^3}{ca} + \frac{c^3}{ab}$$

Ex. 2: Let A, B and C be the magnitude (in radians) of the three angles of a triangle, and a, b and c be the lengths of the sides opposite to these angles respectively. Prove

that
$$\frac{aA + bB + cC}{a + b + c} \geq \frac{\pi}{3}$$

Ex. 3: Let a, b, c be positive real numbers. Prove that $a^{2a}b^{2b}c^{2c} \geq a^{b+c}b^{c+a}c^{a+b}$.

Ex. 4: Let a, b, c be positive real numbers. Prove that $\frac{a}{2a+b+c} + \frac{b}{a+2b+c} + \frac{c}{a+b+2c} \leq \frac{3}{4}$.

Ex. 5: (20th IMO (1978) No. 5)

If $a_1, a_2, a_3, \dots, a_n$ are n positive integers, all of them unequal. Prove that

$$\frac{a_1}{1^2} + \frac{a_2}{2^2} + \frac{a_3}{3^2} + \dots + \frac{a_n}{n^2} \geq 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}.$$

Ex. 6: For any sequence of positive real numbers $\{a_i\}$, and positive integers α and β , prove

that
$$\frac{\sum_{i=1}^n (a_i)^{\alpha+\beta}}{n} \geq \left(\frac{\sum_{i=1}^n (a_i)^\alpha}{n} \right) \left(\frac{\sum_{i=1}^n (a_i)^\beta}{n} \right)$$

Ex. 7: Let x, y, z be positive real numbers such that $xyz = 1$, prove that

$$\frac{x^3}{(1+y)(1+z)} + \frac{y^3}{(1+z)(1+x)} + \frac{z^3}{(1+x)(1+y)} \geq \frac{3}{4}.$$

Ex. 8: If a, b, r are positive real numbers such that $a + b = 1$, prove that

$$\left(a + \frac{1}{a} \right)^r + \left(b + \frac{1}{b} \right)^r \geq \frac{5^r}{2^{r-1}}$$

(Note: $f(x) = \left(x + \frac{1}{x} \right)^r$ is a convex function for $0 < x < 1$.)

Ex. 9: P is a point inside $\triangle ABC$. Prove that among the angles PAB, PBC and PCA , at least one of them is less than or equal to 30° .

(Note: $f(x) = \log_e (\sin x)$ is a concave function for $0 < x < \pi$.)

Ex. 10: Let a_i ($i = 1, 2, \dots, n$) be positive real numbers such that $a_1 + a_2 + \dots + a_n = 1$.

For any non-negative real numbers x_i ($i = 1, 2, \dots, n$), prove that

$$\sum_{i=1}^n \frac{a_i}{1+x_i} \leq \frac{1}{1+(x_1^{a_1})(x_2^{a_2})\dots(x_n^{a_n})},$$

and the equality holds only when $x_1 = x_2 = \dots = x_n$.

(Note: $f(t) = \frac{1}{1+e^t}$ is a concave function for $-\infty < t \leq 0$.)

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